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A comprehensive review of deep learning-based hyperspectral image reconstruction for agri-food quality appraisal

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Abstract
Hyperspectral imaging (HSI) has recently emerged as a promising tool for various agricultural applications. However, high equipment cost, instrumentation complexity, and data-intensive nature have limited its widespread adoption. To overcome these challenges, reconstructing hyperspectral data from simple, cost-effective color or RGB (red-green-blue) images using advanced deep learning algorithms offers a practically attractive solution for a wide range of applications in food quality control and assurance. Through advanced deep learning algorithms, it is possible to capture and reconstruct spectral information from simple, cost-effective RGB imaging to create a reliable, efficient, and scalable system with accuracy comparable to dedicated, expensive HSI systems. This review provides a comprehensive overview of recent advances in deep learning techniques for HSI reconstruction and highlights the transformative impact of deep learning-based hyperspectral image reconstruction on agricultural and food products and anticipates a future where these innovations will lead to more advanced and widespread applications in the agri-food industry.

Keywords Hyperspectral image reconstruction · Image analysis · Deep learning · Agri-food quality

1 Introduction

Hyperspectral imaging (HSI) is an advanced technology that combines a spectrometer and a camera sensor to capture detailed spatial and spectral data. This technique helps detect different chemical and physical properties in food and related products (He and Sun 2015; Wang et al. 2021a; Sun et al. 2024; Ahmed et al. 2024a). HSI captures and processes images

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that offer detailed information about the spectral properties of a scene or object. Unlike standard imaging techniques, HSI collects images over a much broader range of narrow spectral bands, often spanning from tens to hundreds of bands (Kamruzzaman 2023). This wide range of spectral bands enables the identification of specific information about the chemical composition, material properties, and other physical characteristics of food in an image (Elmasry et al. 2012). A key advantage of HSI is its non-invasive nature, which enables the examination of food materials without causing any damage or change (Kamruzzaman et al. 2015; Sim et al. 2024). Additionally, HSI is also valuable in environmental monitoring, such as detecting changes in vegetation, assessing forest health, or monitoring land-use alterations (Lee et al. 2024). It has practical applications in food industries for quality control, like identifying defects or contaminants in food products (Ordoudi et al. 2014; Wu et al. 2023; Sunil et al. 2023). Overall, these benefits demonstrate the versatility of HSI, making it a valuable tool for a range of potential applications in the agriculture and food sectors (Mahesh et al. 2015; Lu et al. 2020; Khan et al. 2022; Wang et al. 2023; Esmaeily et al. 2024).

Despite its vast potential, the adoption of HSI technology faces several challenges (Li et al. 2017; He et al. 2022; Khan et al. 2022; Zhang et al. 2022). Traditional HSI systems, often based on spatial and spectral scanning methods like push-broom or variable-filter systems, are generally bulky and slow. They also feature moving parts and require long exposure times (Arad et al. 2020). Additionally, the equipment used to capture hyperspectral images is typically expensive and complex (Gao et al. 2010; Salazar-Vazquez and Mendez-Vazquez 2020). Even with high-cost systems, image acquisition is time-consuming, a significant barrier to rapid, automated quality assessments in the agri-food sector. These systems also struggle with lower spatial resolution compared to conventional imaging methods, making it difficult to identify individual features within an image (Mei et al. 2022). Further, achieving high spectral resolution often requires sacrificing spatial and temporal resolution, limiting its effectiveness in fields requiring detailed imaging, such as medical and agri-food monitoring (Villa et al. 2010; Fang et al. 2018). In scenarios where light is insufficient, the signal-to-noise ratio (SNR) of hyperspectral data presents an additional challenge. A low SNR can lead to unreliable data, affecting the accuracy of analyses and interpretations. This is particularly problematic in remote sensing, where signals from target objects are often weak (Chang 2022). Furthermore, managing and analyzing the substantial amount of data generated by HSI requires extensive computational resources and expertise, and the technology's sensitivity to environmental variations can affect data accuracy (Pu et al. 2015, 2023; Wendel and Underwood 2017). Additionally, hyperspectral imaging faces challenges like multicollinearity and incomplete or noisy data, complicating product characterization (Wambugu et al. 2021). While recent developments in snapshot HSI technologies have improved the potential for real-time spectral image acquisition, these systems often compromise on spatial and spectral resolution. Finding a balance between performance and practicality in HSI systems still remains a crucial focus of ongoing research and development.

The high cost, complexity of equipment, low spatial resolution and limited image availability have all hindered the widespread adoption of traditional HSI systems. To overcome these limitations, there is growing interest in developing methods to reconstruct hyperspectral data from standard RGB images (Li et al. 2017; Arad et al. 2020; Zhao et al. 2020b; Zhang et al. 2022). RGB imaging provides excellent spatial resolution at a low cost, making it widely used in food, postharvest quality assessment, and agricultural applications (Mei

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et al. 2022). RGB images offer rich texture and structural information useful for object recognition and scene analysis, but they lack the spectral depth of HSIs. This spectral depth is crucial for identifying the physical and chemical attributes of food samples like fruits, vegetables, and animal-derived foods. Therefore, converting readily available RGB images into accurate hyperspectral images emerges as a promising strategy, merging high spatial resolution with comprehensive spectral data, particularly beneficial in agri-food applications, from lab experiments to field operations (Fu et al. 2020; Ilniyaz et al. 2023; Gkillas et al. 2023). This method represents a more accessible and less expensive alternative that drastically reduces the complexity of hyperspectral technology applications. Hyperspectral image reconstruction methods were initially very complex (Fu et al. 2021), but the advent of deep learning has opened new possibilities in this field (Ahmed et al. 2023b; Archana and Jeevaraj 2024). Algorithms now show impressive results in reconstructing hyperspectral images from even low-quality RGB inputs, effectively combining spatial clarity with essential spectral details (Xie et al. 2019; Qin et al. 2023). These advancements make hyperspectral data more accessible and easier to use, helping various sectors make more accurate decisions (Zhao et al. 2022; Cui et al. 2022; Deng et al. 2024). Deep learning's proficiency in parsing complex patterns in extensive datasets allows for superior-quality reconstructions from incomplete or noisy data (Tian et al. 2020; Wang et al. 2021b). However, transforming RGB to hyperspectral data is complex, as it involves reconstructing detailed spectral information from the limited data in RGB values, a challenging inverse problem (Shi et al. 2018; Pan et al. 2023). Nonetheless, this approach has tremendous potential, especially in agri-food applications like precision agriculture and quality assessment of harvested produce. It allows precise identification and quantification of chemical compounds and other quality parameters in food samples. Initiatives like the NTIRE (New Trends in Image Restoration and Enhancement) spectral reconstruction challenges have catalyzed progress in developing these algorithms (Arad et al. 2020). However, fully integrating these algorithms is still an underexplored area in agriculture and food. Because consumer cameras are widely available and high-quality color images are easy to obtain at low cost, successfully applying these techniques could significantly expand the use of HSI.

As a relatively new area of study, there are limited reviews in existing literature on HSI reconstruction from RGB images. Zhang et al. (2022) explored methods for hyperspectral reconstruction, including prior-based and data-driven approaches. Chen et al. (2023) offered a comprehensive look at deep learning techniques for hyperspectral super-resolution. While the reviews provided a broad overview of the reconstruction algorithms, they did not adequately address how these algorithms are being applied to specific domains. Despite its huge potential in the agri-food sector, there has been no published review specifically dedicated to image reconstruction within this field. Therefore, the main objectives of this review paper are to address and focus valuable insight into hyperspectral image reconstruction using deep learning-based techniques for applications in the agri-food domain towards contributing to global food security and food quality and safety issues through advanced data processing and technological advancement.

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2 Principles of HSI technology

HSI was originally developed by NASA for remote sensing applications in space exploration and earth observation. Over time, HSI technology has evolved and expanded into various fields, including agriculture (Wang et al. 2021a, 2023), food quality evaluation (Ahmed et al. 2023a; Sun et al. 2024), medical imaging (Zhang et al. 2024), environmental monitoring (Peyghambari and Zhang 2021) and so on (Ahmed et al. 2023a; He and Sun 2015). The technology combines imaging and spectroscopy in one system to simultaneously capture both spatial and spectral information and generates a complete spectrum for each pixel in an image (ElMasry and Sun 2010). Unlike traditional RGB imaging, which captures only color bands, HSI acquires hundreds of narrow spectral bands across a range of spectral wavelengths. This creates a 3D data cube or hypercube with two spatial dimensions and one spectral dimension that supports detailed analysis of both spatial and spectral characteristics within a sample. Each material exhibits a distinctive spectral signature, characterized by specific patterns of light absorption, reflection, transmission across different wavelengths (Sarić et al. 2022). These spectral signatures enable the identification and differentiation of materials within each pixel of an image. HSI operates in reflection, transmission, and fluorescence modes, using four main image acquisition methods: point scanning (whiskbroom), line scanning (pushbroom), area scanning (tunable filter), and snapshot imaging. Point scanning captures one pixel at a time and moves across the sample, line scanning captures a line of pixels, area scanning records the entire sample one wavelength at a time, and snapshot imaging captures the entire hypercube in a single shot. Line scanning, or pushbroom, is especially popular in food inspection due to its consistency and efficiency in on-line applications. A typical pushbroom HSI system includes a CCD (Charge-Coupled Device) or CMOS (Complementary Metal Oxide Semiconductor) camera, an illumination unit, a spectrograph with a C-mount lens, and computer software for image acquisition (Kim et al. 2022). Although HSI captures significantly more data than a standard RGB image and thus requires longer acquisition times, its detailed spectral data makes it invaluable for product quality analysis.

Consequently, HSI technology has become a valuable tool in the agri-food domain. With increased expectations for high-quality and safe agri-food products, the need for accurate, fast, and objective quality determination of these characteristics continues to grow. The quality assurance methods used in the food industry have traditionally involved human visual inspection. Such methods are tedious, laborious, time-consuming, and inconsistent. Thus, HSI with all its advantages for non-invasive and non-destructive quality evaluation ability, emerges as the solution to the limitations in the agri-food sector. However, the higher cost and increased complexities hinder the widespread applicability of HSI technology. Therefore, reconstructing HSI from commonly available RGB images could be a more affordable solution, expanding its applications in the agriculture and food sectors.

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3 Fundamentals of HSI reconstruction

3.1 Conversion of hyperspectral images from RGB images

HSI involves capturing a wide range of electromagnetic wavelengths, often spanning tens or hundreds of bands. This capability allows HSI to provide image data and detailed spectral information unique to different objects, making it highly effective in identifying specific materials or conditions. However, traditional HSI systems suffer from low spatial resolution, which limits their utility. In contrast, standard RGB images offer rich spatial detail but contain only three spectral channels, i.e., red, green, and blue, in the visible range (400–700 nm). Due to limited spectral information, RGB images can be used for applications that require spectral information within the visible range. Especially for agricultural products with complex compositions, different bands with spectral properties beyond the visible range are necessary to characterize completely. While RGB imaging techniques can be cost-effective, they cannot be used for many applications due to limited spectral information. To overcome these limitations and utilize the benefits of both RGB and HSI, many research endeavors have focused on hyperspectral image reconstruction from RGB images (Arad et al. 2018, 2020; Zhang et al. 2022). This process utilizes deep learning models to bridge the gap between the simplicity and spatial richness of RGB imaging and the detailed spectral analysis offered by HSI. By converting RGB images into HSI with advanced deep learning algorithms, it becomes possible to accurately estimate the spectral signatures at a pixel-by-pixel level. Such a conversion effectively mitigates the high costs associated with direct hyperspectral image capture while enhancing the information depth beyond what is achievable with standard RGB imaging. The conversion process from RGB to hyperspectral imaging is complex and often framed as an inverse problem. An overview of the deep-learning-based reconstruction from RGB to HSI data is depicted in Fig. 1. Mathematically, this transformation can be represented as follows:

$$\widehat{H}\left(x, y, \lambda\right) = f\left(I_{RGB}\left(x, y, 3\right)\right) \tag{1}$$

where, $I_{RGB}\left(x, y, 3\right)$ denotes the RGB images, characterized by spatial dimensions $x \times y$ and encompassing the red, green, and blue channels. $f\left(.\right)$ represents the deep learning algorithm employed for the reconstruction of the hyperspectral image. $\widehat{H}\left(x, y, \lambda\right)$ is the reconstructed hyperspectral image, retaining the spatial dimensions $x \times y$ and comprising λ spectral channels. The complexity of the conversion increases significantly with the increase of λ. Considering a CNN method is applied for the reconstruction purpose, the function f in Eq. 1 can be further broken down into Eqs. 2–3 below:

$$I_1 = f_1\left(I_{RGB}\right) = activation\left(I_{RGB} * W_1 + B_1\right) \tag{2}$$

$$\widehat{H} = I_n = f_n\left(I_{n-1}\right) = activation\left(I_{n-1} * W_n + B_n\right) \tag{3}$$

here, '*' denotes the convolution operation, W is the weight, B is the bias, and n refers to the number of layers in the network. Finally, the loss is calculated and backpropagation is applied to update the weight and bias in the network for each epoch.

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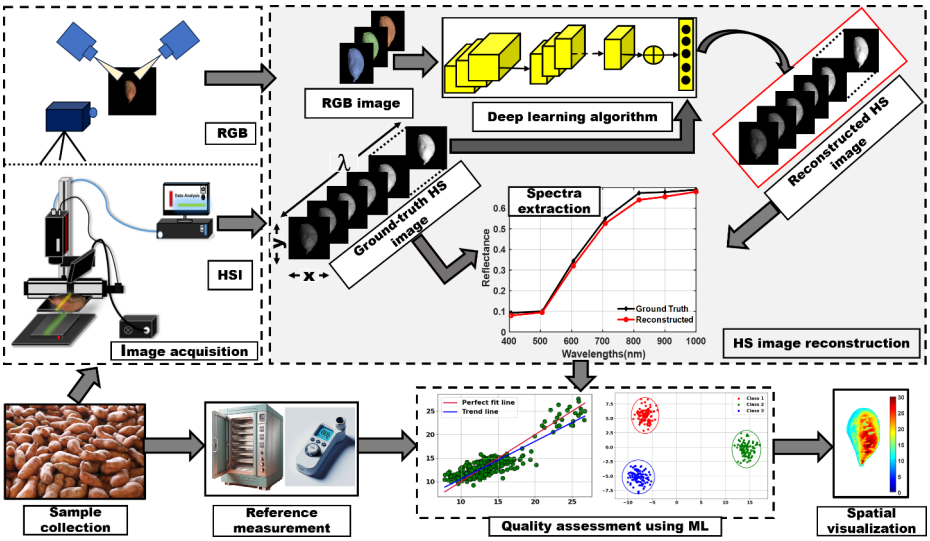


Fig. 1 Overview of deep learning-based hyperspectral image reconstruction to assess the quality of agri-food products

3.2 Performance evaluation metrics

Evaluating the effectiveness of a reconstruction algorithm is crucial to determine its accuracy and efficiency. To assess this performance, several standard metrics are employed. These metrics primarily compare the reconstructed images and spectra with their original, ground-truth counterparts. By measuring the degree of similarity between the reconstructed and original data, these metrics provide a clear indication of the algorithm's performance. The key metrics used for this evaluation are detailed below:

3.2.1 Mean squared error (MSE)

Mean squared error (MSE) is a metric that computes the average of the squared differences between predicted ($\widehat{H}_i$) and actual values (H_i) (Wang et al. 2022). This method is particularly effective for data with a Gaussian distribution, as it disproportionately penalizes larger errors more than smaller ones, ensuring a focus on significant discrepancies.

$$\text{MSE} = \frac{1}{N} \sum {}^{N}_{i=1} (H_i - \widehat{H}_i)^2 \tag{4}$$

3.2.2 Root mean squared error (RMSE)

The Root Mean Squared Error (RMSE) is derived by taking the square root of the MSE. This calculation results in an error measure in the same units as the predicted outputs. RMSE offers a more intuitive grasp of the error magnitude compared to the MSE, as it aligns the error metric more closely with the scale of the actual data (Cai et al. 2022).

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$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(H_i - \widehat{H}_i\right)^2} \tag{5}$$

3.2.3 Mean absolute error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of errors between predicted and actual values, ignoring their direction. It is generally more resilient to outliers compared to MSE, as it does not square the errors, which can disproportionately increase the impact of large outliers (Wang et al. 2022).

$$\mathrm{MAE} = \frac{\sum_{i=1}^{N}|\widehat{H}_i - H_i|}{N} \tag{6}$$

3.2.4 Mean relative absolute error (MRAE)

Mean Relative Absolute Error (MRAE) is a metric for assessing the average absolute discrepancies between actual and forecasted values, expressed as a proportion of the actual values (Cai et al. 2022).

$$\mathrm{MRAE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|\widehat{H}_i - H_i|}{H_i} \tag{7}$$

3.2.5 Back propagation mean absolute error (BPMRAE)

Back propagation mean absolute error (BPMRAE) is used to calculate the colorimetric accuracy (Zhao et al. 2020b).

$$\mathrm{BPMRAE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|(R\times\widehat{H})_i - (R\times H)_i|}{H_i} \tag{8}$$

3.2.6 Peak signal-to-noise ratio (PSNR)

The peak signal-to-noise ratio (PSNR) measures the ratio between the maximum possible value of the original signal such as an image sample and the maximum noise level, which is the difference between the original and a predicted (or compressed) version of the signal (Hore and Ziou 2010). In simpler terms, PSNR quantifies how much the quality of the original signal exceeds the errors introduced, such as during compression or transmission. A higher PSNR indicates a closer match to the original signal, suggesting higher quality with less noticeable errors.

$$\mathrm{PSNR} = 10.\log_{10}\left(\frac{MAX_I^2}{MSE}\right) \tag{9}$$

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3.2.7 Structural similarity index (SSIM)

The Structural Similarity Index (SSIM) measures the similarity between two images by analyzing their luminance, contrast, and structural information (Hore and Ziou 2010). It evaluates how variations in these elements influence the perceived quality of the images, going beyond simple pixel-by-pixel comparisons. A higher SSIM score signifies a greater resemblance between the images, reflecting a minimal loss in perceived image quality.

$$\mathrm{SSIM} = \frac{(2\mu_{H}\mu_{\widehat{H}} + C_1)(2\sigma_{H\widehat{H}} + C_2)}{(\mu^{2}_{H} + \mu^{2}_{\widehat{H}} + C_1)(\sigma^{2}_{H} + \sigma^{2}_{\widehat{H}} + C_2)} \tag{10}$$

3.2.8 Spectral angle mapper (SAM)

The Spectral Angle Mapper (SAM) technique evaluates spectral similarity by measuring the angle between spectra from a sample and an image as vectors. This angle quantifies how closely the two spectra match, indicating similarity in material composition. A smaller angle suggests a high degree of similarity, implying a good match between the sample and image properties (Han et al. 2018a).

$$\mathrm{SAM} = \cos^{-1}\left(\frac{\sum_{i=1}^{b}\widehat{r}_i r_i}{\sum_{i=1}^{b}\widehat{r}_i^2 \sum_{i=1}^{b} r_i^2}\right) \tag{11}$$

where, $\widehat{H}$ = Reconstructed Hyperspectral Image, H=Ground Truth Hyperspectral Image, R = Response function, N = The number of images, $\mathrm{M}ax$ = Maximum possible pixel value, μ_H = Luminance, σ_H = Signal Contrast, C_1 = Constant, $\widehat{r}$ = Reflectence of the generated image, r = Reflectence of the reference image, b = Number of total defined bands.

3.3 Deep learning-based reconstruction algorithms

Hyperspectral image reconstruction, an emerging field of scientific research, has seen significant advancements through efforts like the NTIRE spectral reconstruction challenges (Arad et al. 2020, 2022). These challenges, which focus on developing algorithms for converting RGB images into hyperspectral images, utilize benchmark datasets such as NTIRE-2018 (Arad et al. 2018), NTIRE-2020 (Arad et al. 2020), NTIRE-2022 (Arad et al. 2022), iCVL (Arad and Ben-Shahar 2016), CAVE (Yasuma et al. 2010), NUS (Nguyen et al. 2014), Harvard (Chakrabarti and Zickler 2011), and KAUST-HS (Li et al. 2021) to evaluate and enhance algorithm performance. Each dataset provides a unique test environment with varying spectral bands (400–730 nm), resolutions, and scenarios, ranging from the progressively expanding NTIRE series to the specific setups of CAVE and Harvard, thereby aiding in the thorough assessment and improvement of spectral reconstruction techniques. Further exploration of these algorithms reveals a categorization into general neural network structures (depicted in Fig. 2), which are detailed in subsequent sub-sections based on their configura-

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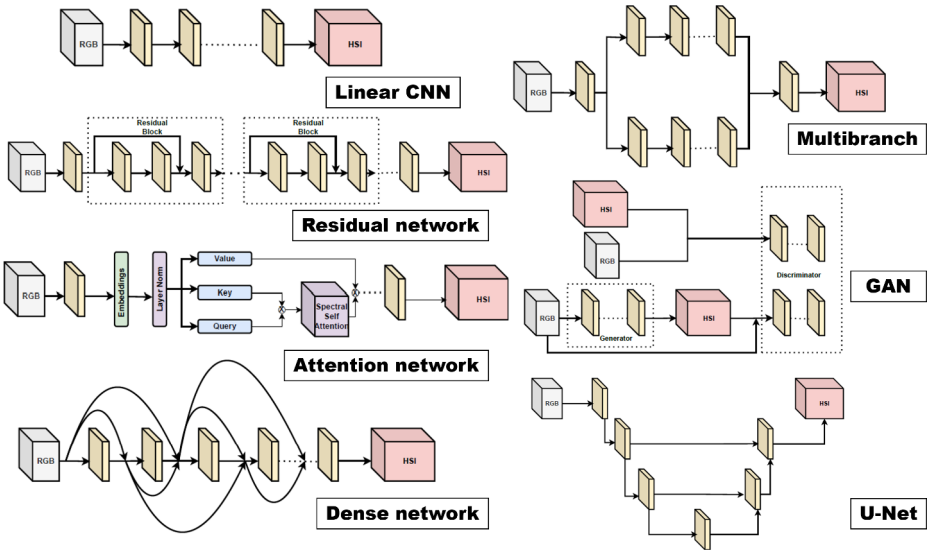


Fig. 2 General structures of deep learning-based image reconstruction algorithms

tion and functionality and the performance of these algorithms on the benchmark dataset are listed in Table 1.

3.3.1 Linear convolution neural networks (CNNs)

3.3.1.1 HSCNN HSCNN, an innovative deep-learning model, leverages the structure of the very deep CNNs for super-resolution (VDSR) architecture, initially designed for enhancing spatial resolution, to transform RGB and compressed-sensor images into hyperspectral data (Xiong et al. 2017). This adaptation builds on VDSR's improvements over earlier models like SRCNN by employing a complex training regime enhanced by a higher learning rate and gradient clipping, facilitating more effective learning. Key to VDSR and by extension, HSCNN is the technique of combining a low-resolution input with the network's output to generate high-resolution images. However, HSCNN diverges in its application: it repurposes VDSR's super-resolution focused architecture for spectral reconstruction, translating RGB images into detailed hyperspectral outputs. This strategic modification showcases HSCNN's unique capability to extend a super-resolution framework to the spectral domain, illustrating its specialized application in hyperspectral image reconstruction.

3.3.1.2 SR2D/3DNet The SR2D/3DNet framework presents two models: one with 2D convolutions for spatial relationships in images and another with 3D convolutions for capturing both spatial and spectral dimensions (Koundinya et al. 2018). This algorithm addresses the challenge posed by the vast channel depth of hyperspectral images, which far exceeds that of RGB images. Experimental results reveal that 3D convolutions slightly improve reconstruction accuracy by integrating spatial and spectral information more effectively than 2D convolutions, which, however, offer nearly comparable accuracy with the advantage of

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Table 1 Performance of the deep learning-based hyperspectral image reconstruction algorithms

Method	Network	Framework	Dataset used	Performance	Agri-food application	References
HSCNN	Linear-CNN	Caffe	iCVL	NRMSE = 0.0398	No	Xiong et al. (2017)
SR2D Net	Linear-CNN	Keras/ TensorFlow	NTIRE-2018	RMSE = 21.394 MRAE = 0.020	No	Koundinya et al. (2018)
			iCVL	RMSE = 2.12 MSE = 0.0086		
			CAVE	RMSE = 3.05 MSE = 0 0.0075		
SR3D Net	Linear-CNN	Keras/ TensorFlow	NTIRE-2018	RMSE = 20.010 MRAE = 0.018	No	Koundinya et al. (2018)
			iCVL	RMSE = 1.115 MSE = 0.0051		
			CAVE	RMSE = 2.86 MSE = 0.0065		
Residual HSRCNN	Residual-CNN	Caffe	CAVE	RMSE = 4.78 SAM = 7.37 PSNR = 35.07	No	Han et al. (2018b)
			HARVARD	RMSE = 3.33 SAM = 5.31 PSNR = 38.93		
HR-ResNet	Residual-CNN	Tensorflow	Rice Dataset	RMSE = 1.20 MAE = 0.61	Yes	Lailyshofa and Saputro (2023)
SREffi-cientNet	Residual-CNN		ICVL	RMSE = 0.6324	No	Can and Timofte (2018)
			CAVE	RMSE = 2.613		
			NUS	RMSE = 2.83		
EDSR	Residual-CNN	Pytorch	Div2K	PSNR = 35.12 SSIM = 0.9699	Yes	Lim et al. (2017)
HRNet	Attention	Pytorch	NTIRE-2020 HS Track-1 (Clean)	MRAE = 0.0323	Yes	Zhao et al. (2020c)
			NTIRE-2020 HS Track-2 Real-World	MRAE = 0.0620		
AWAN	Attention	Pytorch	NTIRE-2020 (Clean)	MRAE = 0.0301	Yes	Li et al. (2020)
			NTIRE-2020 Real-World	MRAE = 0.0621		
MST++	Attention	Pytorch	NTIRE 2022 Val	MRAE = 0.1645 RMSE = 0.0248	Yes	Cai et al. (2022)
			NTIRE 2022 Test	MRAE = 0.1131 RMSE = 0.0231		
Restormer	Attention	Pytorch	NTIRE 2022 Val	PSNR = 33.40 RMSE = 0.0274 MRAE = 0.1833	Yes	Zamir et al. (2022)
PADUT	Attention	Pytorch	CAVE	PSNR = 38.89 SSIM = 0.9740	No	Li et al. (2023)
SPECAT	Attention	Pytorch	CAVE	PSNR = 40.37 SSIM = 0.9860	No	Yao et al. (2024)

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Table 1 (continued)

Method	Network	Framework	Dataset used	Performance	Agri-food application	References
CESST	Attention	Pytorch	NTIRE-2022	PSNR = 35.14 MRAE = 0.1497	No	Yang et al. (2024b)
			iCVL	PSNR = 33.25 MRAE = 0.1230		
HSCNN-r	Residual-CNN	Pytorch	NTIRE-2018 (Clean)	MRAE = 0.0139 RMSE = 13.987	Yes	Shi et al. (2018)
			NTIRE-2018 Real-World	MRAE = 0.0321 RMSE = 25.014		
HSCNN-d	Dense Network	Pytorch	NTIRE 2018 (Clean)	MRAE = 0.0138 RMSE = 14.452	Yes	Shi et al. (2018)
			NTIRE 2018 Real-World	MRAE = 0.0310 RMSE = 24.067		
MPRNet	Multi-Branch	Pytorch	Synthetic datasets	PSNR = 32.73 SSIM = 0.921	Yes	Zamir et al. (2021)
SRLWRD-Net	Multi-Branch	Keras	NTIRE 2020 HS Track-1	MRAE = 0.05478	No	Nathan et al. (2020)
HINet	Multi-Branch	Pytorch	Deraining dataset	PSNR = 33.03 SSIM = 0.926	Yes	Chen et al. (2021)
SRPFMNet	Multi-Branch	Pytorch	NTIRE 2018	RMSE = 1.03 PSNR = 49.29 SAM = 1.05 SSIM = 0.9955	No	Zhang et al. (2020)
			CAVE	RMSE = 4.54 PSNR = 36.33 SAM = 7.07 SSIM = 0.9611		
			HARVARD	RMSE = 2.54 PSNR = 41.54 SAM = 3.76 SSIM = 0.9796		
SRUNet	U-Net	Pytorch	Urban Dataset	PSNR = 26.22 SSIM = 0.7902	No	Feng et al. (2022)
SAPWNet	GAN	Pytorch	iCVL	RMSE = 1.445 PSNR = 32.532 SSIM = 0.932	No	Liu and Zhao (2020)

NRMSE normalized root mean squared error, *MAE* mean absolute error, *MRAE* mean relative absolute error, *MSE* mean squared error, *RMSE* root mean squared errors, *SAM* spectral angle mapper: *BPMRAE* back-projection mean relative absolute error, *PSNR* peak signal-to-noise-ratio, *SSIM* structural similarity index.

faster processing. Both architectures are built on a simple structure of convolutional layers followed by nonlinear activation functions, emphasizing efficiency and effectiveness in spectral reconstruction. The loss function of the algorithm is calculated using Eq. 12.

$$\text{Loss} : \left\| H - \widehat{H_L} \right\|_2^2 \tag{12}$$

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3.3.2 Residual networks

3.3.2.1 Residual hyper-spectral reconstruction CNN (residual HSRCNN) The Residual Hyper-Spectral Reconstruction CNN (Residual HSRCNN) (Han et al. 2018b) introduces a cutting-edge CNN-based technique for enhancing the spectral resolution of hyperspectral images from RGB inputs, targeting the hyperspectral image super-resolution challenge. This approach, known as HSRCNN, harnesses convolutional neural networks to capture the unique spectral characteristics of various materials within an image. It excels in aligning the reconstructed high-resolution hyperspectral images closely with their original high-quality counterparts, maintaining this accuracy during training. The method generates a residual image highlighting the details not captured initially, which is then processed by the same CNN architecture to refine and recover these intricate high-frequency elements. This innovative cycle culminates in the Residual HSRCNN framework, marking a significant advancement in hyperspectral image reconstruction. The loss function of HSRCNN is outlined in Eq. 13.

$$\left\{\widehat{Z}, \widehat{Z}^{Res}\right\} = arg \min_{\varnothing, \varnothing^{Res}} \omega_1 \|Z - f_{HSRCNN}(Y, \theta)\|^2 + \omega_2 \left\|Z^{Res} - f_{Res}\left(f_{HSRCNN}(Y, \theta), \theta^{Res}\right)\right\|^2 \quad (13)$$

where, $Z =$ Output hyperspectral image, $\widehat{Z} =$ Estimated hyperspectral output from the model, $Y =$Input RGB Image.

3.3.2.2 Hyperspectral reconstruction residual net (HR-ResNet) The Hyperspectral Reconstruction Residual Net (HR-ResNet) algorithm incorporates 12 residual blocks, drawing inspiration from the Residual Network (ResNet) design, particularly in its use of shortcut connections to mitigate gradient vanishing issues (Lailyshofa and Saputro 2023). This approach enhances model efficiency by carefully optimizing the number of residual blocks. Stability during the training process is further improved by integrating a batch normalization module at the end of each residual block. By employing these residual blocks and shortcut connections, the architecture achieves enhanced performance, effectively addressing the gradient vanishing problem. The model employs the ReLU activation function and utilizes 64 filters in each of its 12 residual blocks, optimizing both performance and efficiency in hyperspectral image reconstruction. However, the model's practical value remains uncertain, as it has only been tested on the researchers' own dataset.

3.3.2.3 SREffiecientNet The SREfficientNet architecture, leveraging the CNN, is designed to overcome specific challenges in hyperspectral imaging, such as limited data availability and differences in response functions across hyperspectral sensors (Can and Timofte 2018). The study suggests that a relatively shallow CNN architecture can prevent overfitting while successfully capturing the essential transformation from RGB images to the targeted spectral resolution. This approach highlights the balance between model complexity and the ability to accurately interpret and reconstruct hyperspectral data from standard RGB inputs.

3.3.2.4 Enhanced deep residual network (EDSR) The Enhanced Deep Residual Network (EDSR) (Lim et al. 2017) evolves from the SRResNet (Zhao et al. 2020a) model by sim-

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plifying its structure, making the neural network less complex and easier to train. EDSR achieves this by eliminating certain components, such as the batch normalization layers found in SRResNet's and the original ResNet's residual blocks. This modification not only reduces GPU memory usage but also enables the inclusion of additional residual blocks without increasing computational demands. The architecture of EDSR combines basic convolution layers with residual blocks and concludes with an upsampling layer, optimizing the model for efficient performance with lower time and space complexities.

3.3.3 Attention networks

3.3.3.1 HRNet HRNet employs a 4-level hierarchical network, with interconnected networks at each level. The incorporation of distinct levels is purported to amplify information compared to conventional encoder-decoder models (Zhao et al. 2020c). Within each level, convolution operations are followed by residual dense blocks, subsequently succeeded by residual global blocks that utilize MLPs for attention computation, thereby enhancing contextual understanding. To facilitate inter-level fusion, pixelshuffling and pixelunshuffling techniques are employed. While HRNet excels in reconstruction accuracy, it is noteworthy that its computational complexity surpasses that of alternative models. The loss function of the HRNet model can be computed using Eq. 14.

$$\text{Loss} : \sum_{1}^{n} \left\| H - \widehat{H_L} \right\| \tag{14}$$

3.3.3.2 AWAN The AWAN architecture employs a blend of convolution and attention-based modules to enhance image reconstruction by capturing global and local context (Li et al. 2020). Beginning with a basic convolutional layer for initial feature extraction, it progresses to dual residual attention blocks (DRAB) for deeper feature extraction, introducing the AWCA (adaptive weighted channel attention) module to accurately model interdependencies crucial for RGB-to-hyperspectral reconstruction. The AWCA, a squeeze and excitation-based module, learns inter-channel dependencies and provides global information. Additionally, the patch-level second-order non-local (PSNL) module captures distance relationships efficiently in feature maps. The model suggests incorporating camera spectral sensitivity information into the loss function, though its availability varies; nevertheless, even without it, AWAN's performance remains competitive compared to other models.

3.3.3.3 MST++ MST++marks a significant progression in hyperspectral reconstruction technology by tackling two major issues: the need to capture wide-ranging long-distance dependencies and to utilize self-similarity patterns (Cai et al. 2022). This transformer-based model offers a computationally lighter alternative to existing models, addressing challenges that arise with the direct application of transformers to image reconstruction tasks. Unlike traditional transformers, which are adept at handling spatial interactions but struggle with the spatially sparse and spectrally rich nature of hyperspectral images, MST++innovates by focusing on spectral features. It introduces Spectral-wise Multi-head Self-Attention

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(S-MSA), which treats spectral features as tokens to compute self-attention within the spectral domain, bypassing the spatial complexity. This approach, embodied in the self-attention blocks (SABs) arranged in a U-shaped configuration, allows MST++to efficiently extract and leverage multi-resolution spectral information, overcoming the inherent computational and spatial challenges of hyperspectral image reconstruction. Output from the S-MSA can be computed using the following formula:

$$S - MSA\,(X) = \left(Concat_{j=1}^{N}\,(head_j)\right) W + f_p\,(V) \tag{15}$$

where, $head_j = V_j A_j$

3.3.3.4 Restormer Originally developed for image restoration, Restormer can be adapted for hyperspectral image reconstruction, benefiting from key design choices that enhance its efficiency and speed (Zamir et al. 2022). Unlike other transformer-based models, Restormer is optimized for handling high-resolution and high-dimensional data due to its lightweight structure and rapid computation capabilities. This model employs an encoder-decoder transformer architecture that comprehensively processes entire high-dimensional images without segmenting them into smaller sections. It features self-attention mechanisms across feature dimensions to analyze inter-channel dependencies. Each transformer block in Restormer utilizes a novel multi-Dconv head transposed attention system, enabling the capture of both local and non-local interactions through query-key feature interplay across channels. Additionally, Restormer incorporates a gated feed-forward network that selectively transmits the most pertinent features, optimizing the flow of information within the network.

3.3.3.5 Pixel adaptive deep unfolding transformer (PADUT) The Pixel Adaptive Deep Unfolding Transformer (PADUT) addresses key challenges in existing hyperspectral image (HSI) reconstruction methods, such as fixed gradient descent, inadequate 3D handling, and ineffective stage interaction (Li et al. 2023). PADUT introduces a pixel-adaptive descent step in the data module to specifically target pixel-level degradation. The prior module incorporates a Non-local Spectral Transformer (NST) to enhance the recovery process by emphasizing the 3D characteristics of HSI. Furthermore, stage interaction is optimized by using the Fast Fourier Transform (FFT) to better capture the diversity of features across stages. This involves transforming encoder and decoder features from the previous layer into the Fourier domain, enabling a fusion process that highlights various frequency attributes. The outputs of the data module can be formulated using Eq. 16.

$$x^{k+1} = z^k + F_\sigma\, \Phi^T \left(\varphi\, \varphi^T\right)^{-1} \left(y - \varphi\, z^k\right) \tag{16}$$

where, $F_\sigma = \frac{1}{1+\mu}$ z^k = input feature, ϕ = physical mask.

3.3.3.6 SPatial-spEctral cumulative-attention transformer (SPECAT) The SPECAT model addresses the limitations of conventional architectures that predominantly enhance per-

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formance by adding multiple self-attention modules, which is unsustainable due to the disproportionate increase in parameters relative to performance gains (Yao et al. 2024). Instead, SPECAT employs Cumulative-Attention-Blocks (CABs) within an efficient hierarchical framework, effectively capturing non-local spatial and spectral details in hyperspectral images. These blocks consist of Hierarchical Spatial Attention (HSA), Mask Attention (MA), and Global Spectral Attention blocks, leveraging spectral correlations to enhance image reconstruction. Notably, CABs facilitate the extraction of spatial information across spectral bands by processing each local spatial feature map as a token and applying self-attention within the spatial domain. Furthermore, SPECAT utilizes a novel Dual-domain Loss Function (DLF), specifically formulated to incorporate the physical model of spectral reconstruction, ensuring enhanced training efficacy and model robustness. This approach marks a significant shift from traditional methods, emphasizing efficiency and accuracy in processing hyperspectral imagery. the proposed Dual-domain Loss Function of the SPECAT model can be computed using Eq. 17.

$$L = ||H_{\text{out}} - H_0|| + ||\sum\nolimits_{\lambda} M \odot H_{\text{out}} - Y_0||_2 \tag{17}$$

Where H_0 and Y_0 are the true values of the object and projection.

3.3.3.7 Hyperspectral image reconstruction via combinatorial embedding of cross-channel spatio-spectral clues (CESST) CESST introduces a novel approach for exploring the interdependencies among hyperspectral bands by examining inter-channel correlations through a combinatorial technique (Yang et al. 2024b). The CESST model incorporates a spatio-spectral attention block alongside a spectrum-fusion attention module, specifically designed to capture long-range interactions and detailed pixel-level information within hyperspectral images. The architecture is grounded in an encoder-decoder framework, where each channel of the input RGB image is processed through distinct feature extraction blocks. These blocks feed into a newly developed spatio-spectral attention block composed of parallel spatial and spectral Multi-Self-Attention (MSA) units. These units concurrently process and integrate spatial and spectral data to promote enhanced cross-dimensional interactions. The subsequent spectrum-fusion attention module strategically synthesizes correlations among the tripartite representations derived from the spatio-spectral attention block, culminating in the reconstructed hyperspectral image output. The loss function used to train the model can be computed using Eq. 18.

$$L = L_{MIX} + \lambda_1 L_{MRAE} \tag{18}$$

where, $L_{MIX} = \sum_{s=1}^{3} [(\widehat{H}^s, H^s)_{mix}]$ *and* $L_{MRAE} = \sum_{s=1}^{3} \left[\frac{\left|H^s - \widehat{H}^s\right|}{H^s}\right]$.

here, $\widehat{H}^s = reconstructed\ HSI,\ H^s = ground\ truth\ HSI.$

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3.3.4 Dense networks

3.3.4.1 HSCNN+ Two advanced CNN-based hyperspectral reconstruction architectures are presented combinedly as HSCNN+ (Shi et al. 2018). Firstly, the development of a residual network named HSCNN-R is undertaken, consisting of a series of residual blocks. The model's superior performance stems from its contemporary architectural design and optimization achieved by eliminating hand-crafted image upsampling as seen in HSCNN. Building upon the promising results of HSCNN-R, another distinct architecture is introduced, wherein the residual blocks are replaced by dense blocks featuring a novel fusion scheme. This architectural evolution gives rise to a new network named HSCNN-D, substantially augmenting the network's structural complexity to attain heightened precision. The empirical results unequivocally highlight the substantial progress our proposed models bring about, advancing the existing state-of-the-art in hyperspectral reconstruction.

3.3.5 Multi branch

3.3.5.1 SRLWRDNet The SRLWRDNet model adopts a dual-network strategy, simultaneously extracting unique features from an input image through parallel processing (Nathan et al. 2020). These features are then merged to improve the quality of image reconstruction. Initially, the image is processed through a coordinate convolutional layer, adding two extra channels to highlight boundary details and reduce internal inconsistencies, standardizing values between -1 and 1. This feature map feeds into two networks: the Dense feature extractor, drawing from DenseNet's densely connected layers to prevent gradient vanishing, and the multi-hierarchical feature extractor, employing a U-Net-like encoder-decoder framework with convolutions and residual dense attention blocks for in-depth feature extraction. The model's loss function, combining L2 and structural loss, ensures the structural integrity of the reconstructed image. This multifaceted approach allows for comprehensive feature extraction across different scales, bolstered by attention modules for enhanced performance. The attention modules can be formulated by:

$$Channel\ Attention\,(C_A) = F_A[\omega_{\,1}\left(\omega_{\,0}\left(\frac{\sum_{k=1}^{n} x_k}{n}\right)\right) + \omega_{\,1}(\omega_{\,0}\,(max\,(x_k))\,)] \quad (19)$$

$$Spatial\ Attention\,(S_A) = F_{conv}\left(\left[\frac{\sum_{k=1}^{n} C_{Ai}}{n}\right] \| max\,(C_{Ai})\right) \quad (20)$$

3.3.5.2 MPRNet MPRNet is designed for image restoration and can be applied to hyperspectral image reconstruction, employing a three-stage process to enhance the output both contextually and spatially. Instead of simply cascading stages, the architecture integrates a supervised attention block between stages, which facilitates progressive learning and preserves important spectral details (Zamir et al. 2021). The first two stages utilize an encoder-decoder structure, preparing the groundwork for detailed reconstruction. The final stage combines the original input with the enhanced image from the second stage, using a special-

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ized Original Resolution Subnetwork (ORSNet) that maintains the original image's resolution. This approach allows MPRNet to merge the detailed aspects of the original image with the improvements from earlier stages, resulting in a highly refined final output.

$$\text{Loss Function}: L = \sum_{i=1}^{3} [l_{char}(X_s, Y) + \lambda l_{edge}(X_s, Y)] \tag{21}$$

where, l_{char} = Charbonnier loss, l_{edge} = Edge loss, X_s = Restored image, Y = Ground truth image.

3.3.5.3 HINet HINet explores the impact of Instance Normalization on low-level vision tasks by introducing the Half Instance Normalization Block (HIN Block), designed to boost image restoration network performance (Chen et al. 2021). This innovation is key to HINet, a robust multi-stage network that incorporates two U-Net-based subnetworks. These subnetworks are seamlessly integrated using a Cross-stage Feature Fusion (CSFF) module and a Supervised Attention Module, ensuring effective feature integration. Each subnetwork consists of a series of HIN Blocks and Res Blocks, with the HIN Block applying instance normalization to only half of the feature map, leaving the rest unchanged. This distinctive approach enables HINet to achieve superior results across various image restoration tasks, marking a significant advancement in the field.

$$\text{Loss Function of HINet}: -\sum_{i=1}^{2} PSNR((R_i + X_i), Y) \tag{22}$$

where, i = The number of subnetworks, X_i = Input of the subnetwork i, R_i = output of the subnetwork i, Y = Ground Truth.

3.3.5.4 SRPFMNet The SRPFMNet model initially upsamples input images using bicubic interpolation and then processes the upsampled images through a novel Function Mixture Block (FMB) module (Zhang et al. 2020). The FMB module is designed to adaptively determine the optimal receptive field size and mapping function for each pixel, utilizing a separation aggregation strategy. At the outset, a basic convolution layer followed by a ReLU activation function extracts primary features. These features are then fed into parallel subnetworks: one subnet calculates mixing weights, while the others, starting with different convolution sizes, extract features across various receptive fields. The outputs from these subnets are combined to form a cohesive aggregated output, representing the FMB's contribution to enhancing the image's detail and texture.

$$\text{Loss Function}: \min_{\theta} \frac{1}{N} \sum_{i=1}^{N} || y_i - f(x_i, \theta) ||_1 \tag{23}$$

where, $f(x_i, \theta)$ = Reconstructed hyperspectral image, y_i = Ground truth hyperspectral image.

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3.3.6 GANs

3.3.6.1 SAPWNet The efficacy of the SAPWNet model comes from its incorporation of the dual self-attention mechanisms, which jointly enables the model to comprehensively comprehend both localized and global contextual information within the dataset (Liu and Zhao 2020). Consequently, SAPWNet achieves significant enhancements in image quality, coherence, and intricate detailing when compared to conventional GANs. These GANs are meticulously crafted to tackle the complex challenges presented by heavily underconstrained inverse problems. The inaugural innovation encompasses the introduction of the scale attention pyramid UNet (SAPUNet), where the U-Net architecture is applied, and bolstered by dilated convolutions to facilitate robust feature extraction. Embedded within the network, a feature pyramid is established, and an integrated attention mechanism heightens feature selection effectiveness. The conspicuous performance amelioration demonstrated by this model can be attributed to its contemporary architectural design and adeptness in capturing spatial semantics. The SAPWNet extends the U-Net configuration by incorporating an additional branch dedicated to boundary supervision to further enhance the precision of solutions.

$$\text{Total Loss}: L^{*}(G,D) = L_{adv} + \lambda L_1(G) = L_{adv} + \lambda \underbrace{E}_{l_{rgb,ls} \sim P_{data}\left(l_{rgb,l_{hs}}\right)} [\|l_{hs} - G(l_{rgb})\|] \tag{24}$$

I_{hs} = Original hyperspectral image, I_{rgb} = Original input RGB image, λ = Scaler weight to balance loss terms, $P_{data}(l_{rgb,}l_{hs})$ = Data distribution, $P_{model}(l_{rgb})$ = Model distribution.

3.3.7 U-Nets

3.3.7.1 SRUNet The Stacked Reversed U-Shape Network (SRUNet) (Feng et al. 2022) evolves from the U-Net architecture, aiming to improve multi-scale feature extraction and utilization at different resolutions. SRUNet integrates several key components: Shallow Feature Extraction (SFE) for initial processing, the Stacked Reversed U-shape module (SRUM) for iterative feature extraction, the Multi-Scale Backward Fusion Module (MSBFM) for combining features across scales, and the Feature Refinement Module (FRM) for final adjustments. Differing from traditional models that upscale feature maps only at the end, SRUNet's approach through SRUM allows for the capture of intricate features progressively. It employs scale-wise dense connections and backward fusion to enhance multi-scale feature integration, followed by refinement for high-quality image reconstruction. Notably, SRUNet omits pooling layers, typically used in U-Net, to prevent information loss, crucial due to the small datasets characteristic of hyperspectral imaging. An initial preprocessing convolutional layer addresses noisy inputs, with MAE serving as the loss function, tailored for the precise demands of hyperspectral image reconstruction.

These data-driven methods showcase the versatility of machine learning and statistical approaches in hyperspectral image reconstruction, enabling the extraction of meaningful infor-

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mation from complex hyperspectral datasets. The choice of method often depends on the specific goals of the reconstruction task and the characteristics of the hyperspectral data being analyzed

4 Application in quality assessment of agri-food products

The potential of color images, such as RGB images, in evaluating the quality of agricultural products is limited by their narrow range of band information. Hence, HSI reconstruction represents a notable advancement in the field of agricultural and food sciences. This specialized branch of HSI technology is dedicated to creating high-quality spectral images, providing a more in-depth analysis of critical quality parameters such as moisture content, nutrient levels, defects and the presence of contaminants in agri-food products. These insights are crucial for ensuring the safety and overall quality of food. Notably, HSI reconstruction offers a more resource-efficient approach compared to traditional HSI methods, proving particularly beneficial for the agricultural commodities grading industry. Moreover, the ability of HSI reconstruction to work with incomplete data opens new frontiers in real-time monitoring and quality control in the food industry. It allows for the rapid assessment of food quality on production lines, where speed and efficiency are crucial, without the need for extensive and time-consuming data collection processes.

Image reconstruction has shown great promise and found extensive use in fields such as medicine (Wang et al. 2020; Qiu et al. 2023) and remote sensing (Deng et al. 2021; Gao et al. 2021; Hu et al. 2023). Deep learning-based reconstruction algorithms have played a crucial role in enhancing diagnostic accuracy and efficiency, making medical care more reliable and intelligent (Qiu et al. 2023). HSI reconstruction is also utilized in geographical and marine applications. Specifically, reconstructing images in the near-infrared (NIR) region provides significantly richer information for marine studies (Hu et al. 2023). However, the application of image reconstruction in agriculture and food sciences is still in the early stages, primarily confined to academic research and pilot projects. Some challenges need to be addressed, including improving the accuracy of reconstruction algorithms, integrating these systems within existing agricultural and food processing operations, and training staff to effectively use and interpret HSI data. The application of HSI reconstruction in agri-food products is still relatively new and underexplored, as evidenced by the limited number of studies (Zhao et al. 2020b; Lailyshofa and Saputro 2023; Ahmed et al. 2024b). Table 2 provides a comprehensive summary of these research efforts, focusing on the application of HSI reconstruction techniques for quality prediction in agri-food products.

The pioneering work in applying HSI reconstruction to agri-food samples was conducted by Zhao et al. (2020b). Their study implemented the HSCNN-R algorithm to evaluate various parameters, including soluble solid content (SSC), total titratable acidity (TTA), and lycopene content in 33 tomato samples. The outcomes, characterized by an RMSE of 6.22, MRAE of 0.05, and SAM of 1.68, highlighted the reconstruction algorithm's training performance. However, the study also identified significant discrepancies between the reconstructed and ground-truth (GT) spectral data within the 380–740 nm range. This discrepancy led to less accurate predictions for SSC and TTA (listed in Table 2), underscoring the challenges in achieving precise HSI reconstruction for these quality parameters. Fu et al.(2022) used HSI reconstruction techniques in the visible range (400–700 nm) to detect pest-infected disease in maize. They reported an MRAE of 0.07 and RMSE of 0.12

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Table 2 Application of HSI reconstruction in quality evaluation of agri-food products

Product	Reconstruction method used	Spectral range (nm)	Quality parameter	Performance	References
Tomato	HSCNN-R	400–1000	SSC	$R^2 = 0.51$	Zhao et al. (2020b)
			TTA	$R^2 = 0.61$	
			Lycopene	R^2 =0.92	
Maize	HSCNN+	400–700	Pest disease	Precision=94.27%	Fu et al. (2022)
Rice plants	MST++ (Pre-trained)	400–700	SPAD	$R^2 = 0.40$	Yang et al. (2024a)
			Bending force	$R^2 = 0.37$	
			Height	$R^2 = 0.71$	
			DMC	$R^2 = 0.38$	
Rice grains	HR-ResNet	400–1000	-	MAE=0.61 RMSE=1.20 (for reconstruction only)	Lailyshofa and Saputro (2023)
Sweet potato	HSCNN-D	400–1000	SSC	R_p^2= 0.69, RMSEP=0.91, RPD=1.80	Ahmed et al. (2024b)
Sweet potato	HSCNN-D	400–1000	DMC	R_p^2= 0.84, RMSEP=2.74, RPD=2.51	Ahmed et al. (2024c)
	HRNet			R_p^2= 0.88, RMSEP=2.35, RPD=2.93	
	MST++			R_p^2= 0.65, RMSEP=4.04, RPD=1.70	
Egg	HRNet	400–1000	Embryo mortality classification	Accuracy=81.15%	Ahmed et al. (2024d)
	Restormer			Accuracy=80.64%	
	EDSR			Accuracy=77.42%	
	MST++			Accuracy=74.69%	
Beef	MST++	400–1000	Glutamic acid index	R_p^2= 0.84, RMSEP=0.17, RPD=2.46	Dong et al. (2025)

SSC soluble solid content, *TTA* total titratable acidity, *MAE* mean absolute error, *RMSE* root mean squared errors, R^2= coefficient of determination, R_p^2 = coefficient of determination in prediction, *RMSEP* root mean squared errors in prediction, *RPD* ratio of prediction to deviations

to train the HSCNN+algorithm. Subsequently, Yang et al.(2024a) applied the pre-trained MST++method to develop inversion models for reconstructing the images for assessing different physiological parameters of rice. The authors discuss the potential of HSI reconstruction methods by comparing them with their RGB and multispectral counterparts. Another contribution was made by Lailyshofa and Saputro, who utilized the HR-ResNet algorithm to develop a spectral profile for rice grains (Lailyshofa and Saputro 2023). They reported an RMSE and MAE of 1.20 and 0.61 respectively for training the reconstruc-

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tion algorithm. However, they have outlined plans for future research aimed at predicting the amylose content in rice based on the spectra reconstructed through their initial study. More recently, Ahmed et al.(2024b) have expanded the application of HSI reconstruction by applying the HSCNN-D reconstruction method to analyze the SSC in 141 sweet potato samples. They reported an MRAE of 0.86, RMSE of 0.05, and PSNR of 26.86 dB for training the reconstruction model. Their analysis revealed that while the reconstructed spectral data closely matched the actual data in the visible range, discrepancies were observed in the NIR range (700–1000 nm). Expanding on their foundational work, Ahmed et al.(2024c) extended the scope of HSI reconstruction research to include the analysis of dry matter content (DMC) in sweet potatoes, comparing three different reconstruction methods: HSCNN-D, HRNet, and MST++. The study showcased the reconstructed images and spectra of the sweet potato samples using different reconstructed algorithms in Fig. 3a-b. Their findings indicated that HRNet outperformed the other algorithms by delivering the most accurate reconstructions, with the lowest MRAE (0.07) and RMSE (0.03), and the highest PSNR (32.28 dB). While the HSCNN-D method produced reconstructions that closely matched the GT spectra in the visible spectrum, it fell short in the NIR region, showing significant discrepancies. Conversely, the MST++method exhibited notable variations from the GT spectra across the entire spectral range, as illustrated in Fig. 3b. Hyperspectral image reconstruction has also been applied to advance hatchery and meat research. Ahmed et al.(2024d) used the HRNet, MST++, Restormer, and EDSR algorithms to reconstruct hyperspectral images of eggs from RGB images for the early prediction of chick embryo mortality. The study demonstrated the superior performance of the HRNet model, which produced the most accurate reconstructions (MRAE-0.09, RMSE-0.02, and PSNR-36.79 dB) followed by Restormer, EDSR, and MST++ (as visually depicted in Fig. 4a, b). Dong et al.(2025) applied 11 hyperspectral reconstruction techniques to predict the glutamic acid index in beef. They identified MST++as the most effective reconstruction method with an MRAE of 0.01, RMSE of 0.02, and PSNR of 36.97 dB for reconstructing 31 wavebands. The selection of these 31 wavebands was aligned more with the NTIRE template algorithms used for general reconstruction challenges than with the specific needs of agricultural applications. The lack of focus on application specific wavebands may limit the effectiveness of the reconstruction methods for targeted agricultural uses and highlights the need for customized approaches in future studies.

These cumulative efforts underscore the evolving landscape of HSI reconstruction technology, emphasizing its potential to revolutionize the agri-food sectors. By highlighting the strengths and limitations of various reconstruction methods, these studies pave the way

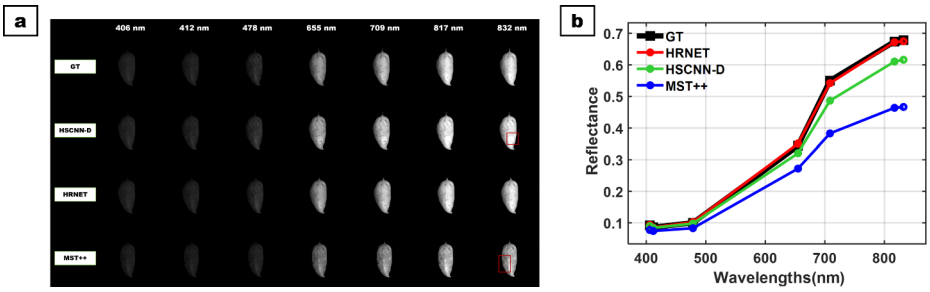


Fig. 3 Ground-truth and reconstructed **a** images and **b** spectra of sweet potato by Ahmed et al. (2024c)

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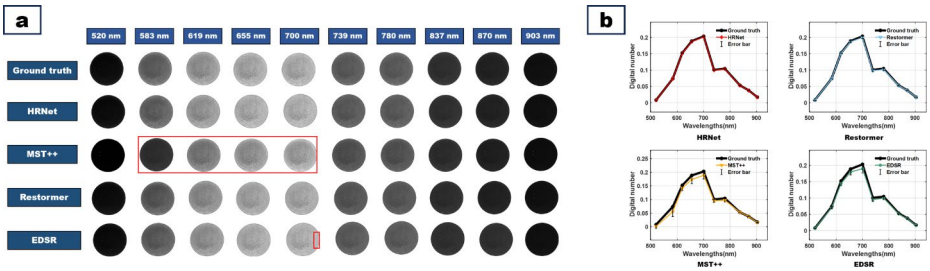


Fig. 4 Ground-truth and reconstructed **a** images and **b** spectra of eggs by Ahmed et al. (2024d)

for further innovation and adoption of HSI technologies, potentially catalyzing significant scientific and industrial advancements in these fields.

5 Opportunity and limitations

Recent advancements in deep learning have significantly expanded their application in predicting agri-food quality, highlighting the role of hyperspectral image reconstruction in this domain. This technology utilizes the detailed spatial and spectral information of hyperspectral data, adeptly overcoming the challenges encountered by traditional HSI systems. The growing significance of this technology is particularly notable in the agricultural and food sectors, where it might offer a paradigm shift in existing methodologies in the coming decade. The fusion of hyperspectral image reconstruction with machine learning has enabled the creation of practical applications, including real-time apps for assessing food quality. The technology could be adapted for smartphones, enhancing computational photography with richer, more detailed information. This technology may also allow agricultural robots to utilize hyperspectral imaging rather than just RGB vision, expanding their functional capabilities. The primary advantage of this technology lies in its ability to facilitate quick, non-destructive quality checks of agricultural products without additional costs or complexities.

The effectiveness of hyperspectral reconstruction largely depends on its capacity to capture spectral signatures from the near-infrared (NIR) region (700–2500 nm). This spectral range is crucial as it provides detailed structural information related to molecular vibrations, which are key in determining the quality attributes of agricultural products (Kamruzzaman et al. 2022). However, most current research focuses on hyperspectral data reconstruction in the visible (400–700 nm) and short-wave infrared (400–1000 nm) spectra. Extending this reconstruction to cover the entire vis-NIR spectrum (400–2500 nm) could greatly enhance applications in food safety, traceability and authenticity verification. Nevertheless, the complexity of this task increases with the number of spectral bands involved. Therefore, strategically selecting key spectral features for reconstruction could be a practical approach to managing this complexity (i.e., product and quality attributes specificity).

While hyperspectral image reconstruction techniques show promise in agri-food systems, completely replacing traditional hyperspectral imaging methods is not straightforward. These deep learning-based techniques face several challenges in their application in the agricultural sector. Usually, deep learning methods need an extensive amount of data to learn the pattern. However, the rapid spoilage of agricultural samples limits the avail-

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ability of data necessary for these techniques. Furthermore, these methods are sensitive to environmental conditions like light and weather, which can lead to inconsistencies in the reconstructed signals (Arad and Ben-Shahar 2016; Mu et al. 2024). In complex scenarios, accurately representing the diversity and variability of the scene can be challenging for hyperspectral image reconstruction. Another significant challenge is the reliance on specific algorithms for image reconstruction. Choosing an inappropriate algorithm can result in inaccurate or biased data. These techniques are also computationally intensive, especially when dealing with large datasets. They often require a trade-off between spatial and spectral resolution, which might compromise the detailed spectral information provided by traditional hyperspectral sensors. Besides, reconstructing hyperspectral images from RGB images is inherently challenging due to the ill-posed nature of the process. This complexity arises because multiple hyperspectral images can be derived from the same RGB image, a phenomenon known as metamerism. Consequently, this can lead to inaccuracies in the reconstructed hyperspectral images, stemming from both the fundamental limitations of RGB images and the complexities involved in solving this type of inverse problem.

Ongoing research and advancements in image reconstruction and sensor technology are essential for overcoming the challenges of HSI reconstruction. Currently, computational limitations restrict its applications, but the development of interactive software could broaden its use across various fields (Ahmed et al. 2024e). Additionally, integrating compressed spectral imaging (Wang et al. 2015) into the RGB-to-HSI reconstruction process can offer a promising solution to enhance reconstruction efficiency. Such advancements could make hyperspectral image reconstruction a more viable alternative to traditional methods in certain situations. In summary, while image reconstruction is a valuable tool for enhancing aspects of image data, it may not completely replace the need for comprehensive hyperspectral image acquisition and processing, especially in applications requiring detailed spectral information. The choice between these methods should be based on the specific needs of the application and the balance between spatial and spectral resolution.

6 Conclusions

Applying deep learning to reconstruct hyperspectral images from RGB images is a significant advancement in agricultural and food sciences. This innovative method has the potential to address traditional challenges of cost, complexity, and portability in hyperspectral imaging technology, making advanced imaging more accessible across various sectors. Deep learning-based reconstruction methods enable detailed spatial and spectral analysis through readily available RGB images. With rapid advancements in deep learning, hyperspectral reconstruction techniques are becoming more mature and efficient. This review highlights the transformative potential of these techniques in revolutionizing agricultural practices, food quality assessment, and postharvest evaluation. Although the application of reconstruction technologies in agriculture is still in its early stages, as these technologies advance, they are expected to support more sophisticated, accurate, and comprehensive applications. This progress will transform agriculture and food engineering, opening new research opportunities and enhancing operational efficiency.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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